Statistics

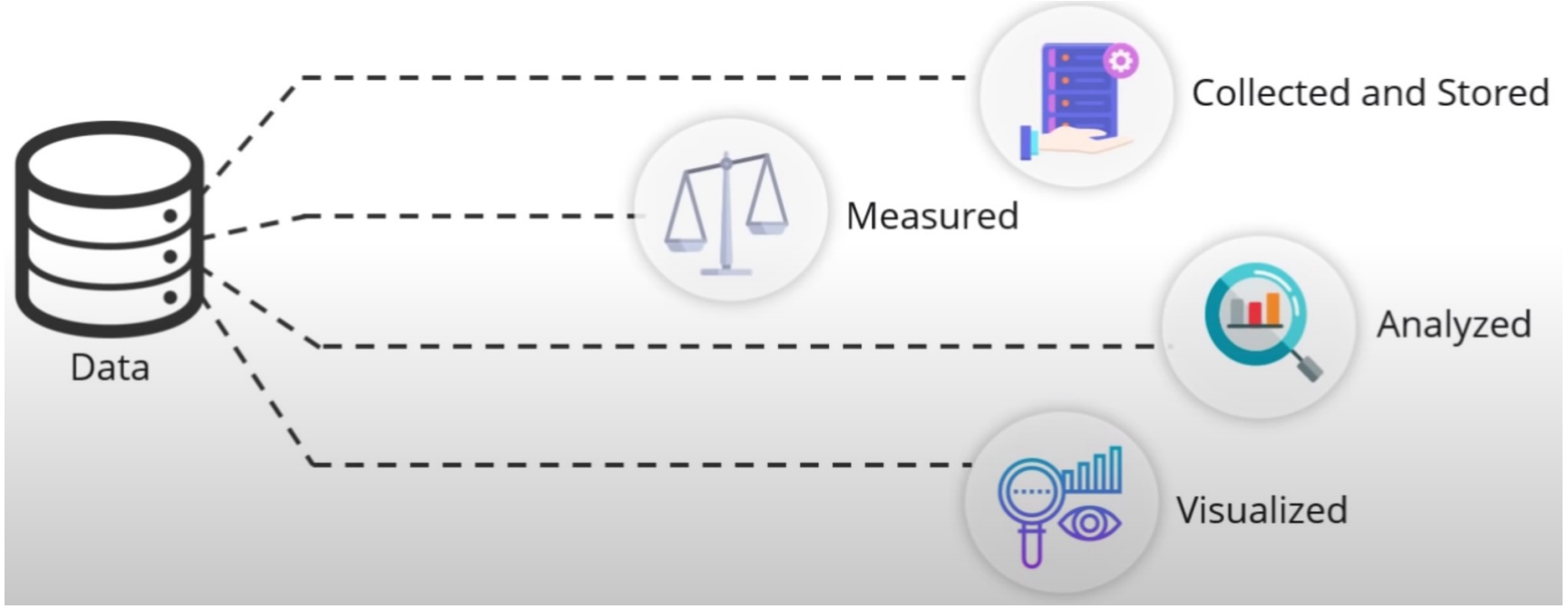
Cao Tien Dung, PhD

Outline

- Probability & Statistics
  - What is data
  - Statistics
  - Distribution
  - Statistical Inference

What is Data?

- Data refers the facts and statistics collected together for reference or analysis.



Categories of Data

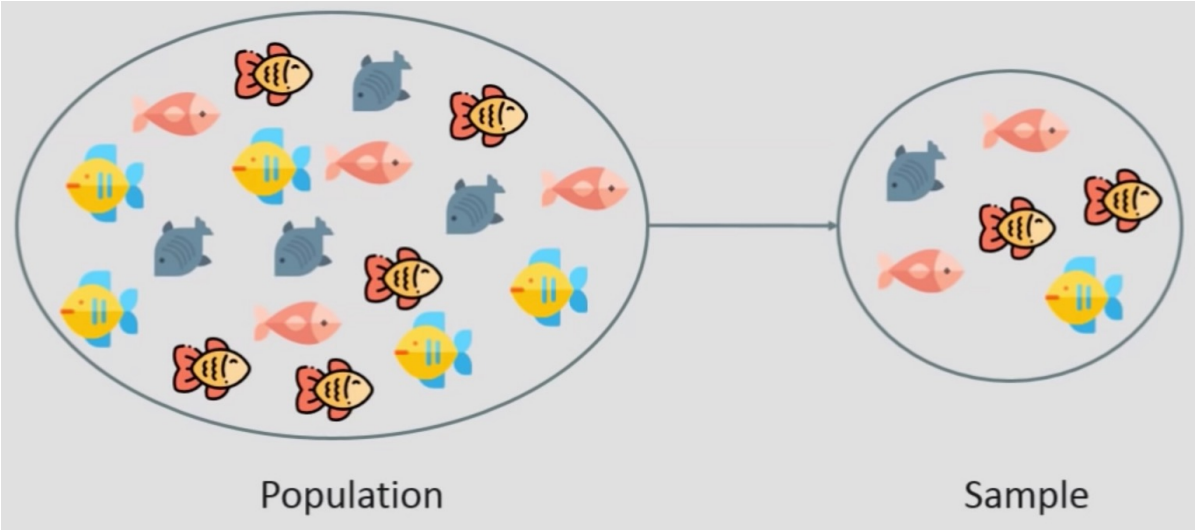
- Qualitative (Categorical): deal with characteristics & descriptors that can't easily measured, but can be observed subjectively.
  - Nominal: data with no inherent order or ranking (e.g., male/female)
  - Ordinal: data with and ordered series (e.g., poor, average, good, excellent)
- Quantitative (Number): deal with number things we can measure objectively.
  - Discrete: hold finite number of possible values (e.g., number of students in class)
  - Continuous: hold infinite number of possible values (e.g., weight of person)

What is Statistics?

- An area of applied mathematics concerned with the data collection, analysis, interpretation and presentation.

Statistics terminologies

- **Population:** The entire group one desires information about
- **Sample:** A subset of the population taken because the entire population is usually too large to analyze. Its characteristics are taken to be representative of the population



Sampling techniques

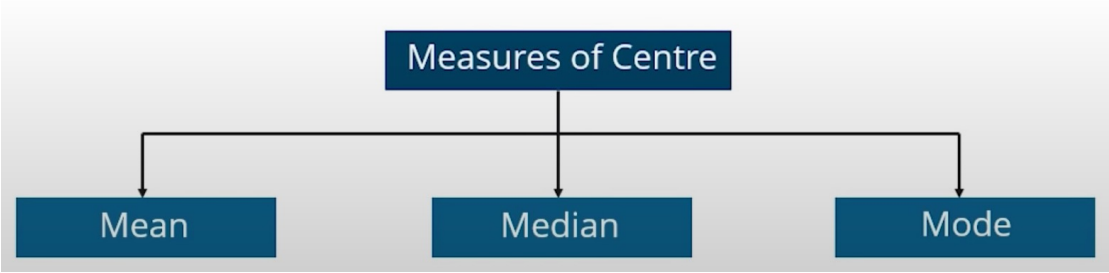
- Probability
  - Random: each member has equal chance of being selected.
  - Systematic: every 2nd record is chosen
  - Stratified: subsets of population are build by sharing at least one common characteristic, and random select from each subset.
- Non-Probability
  - Quota
  - Judgment
  - Convenience

Type of Statistics

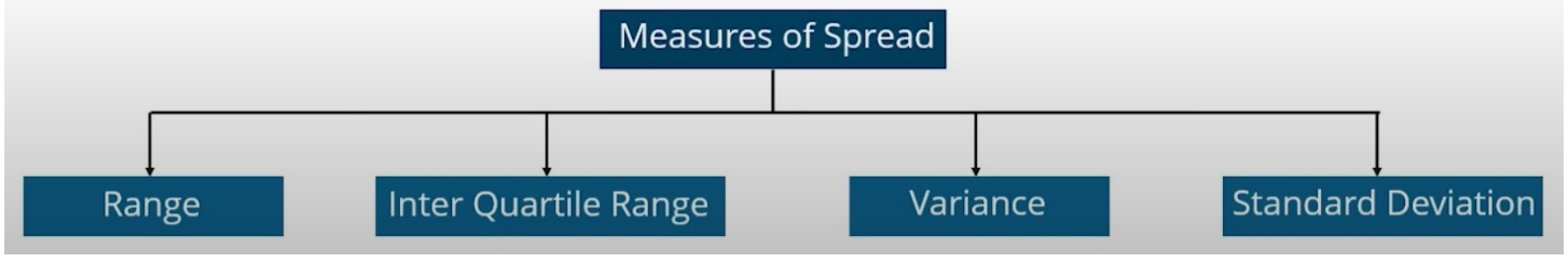
- Descriptive versus Inferential Statistics:
  - Descriptive Statistic: Data description (summarization) such
    - Number data: center, variability and shape.
    - Categorical data: frequency, percentage or proportion
  - Inferential Statistic : drawing conclusion beyond the sample studied (by probability), allowing for prediction.

Descriptive Statistic

- Broken down into 2 categories
  - Measures of Central Tendency



  - Measure of Variability (Spread)

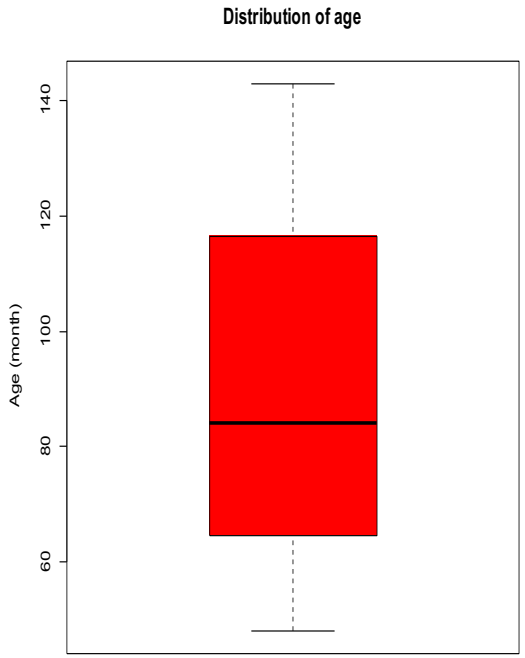


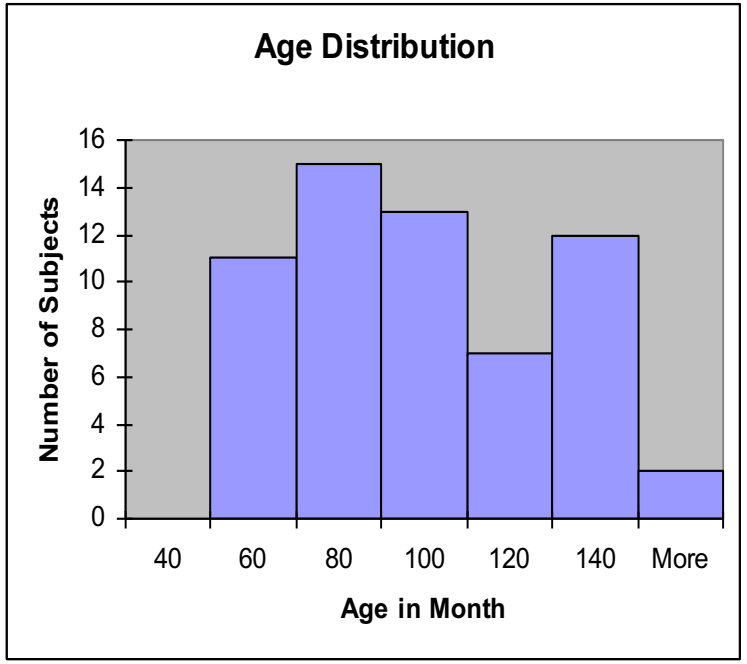
Statistical Description of Data

- Statistics describes a numeric set of data by its
  - Center (mean, median, mode etc)
  - Variability (standard deviation, range etc)
  - Shape (skewness, kurtosis etc)
- Statistics describes a categorical set of data by
  - Frequency, percentage or proportion of each category

How statistics help us

- By simply looking at the data, we fail to produce any informative account to describe the data, however, statistics produce a quick insight into data using graphical and numerical statistical tools





Mean	90.41666667
Standard Error	3.902649518
Median	84
Mode	84
Standard Deviation	30.22979318
Sample Variance	913.8403955
Kurtosis	-1.183899591
Skewness	0.389872725
Range	95
Minimum	48
Maximum	143
Sum	5425
Count	60

Statistics Vocabulary

- Population: the entire group one desires information
- Sample: A subset of the population. Its characteristics are taken to be representative of the population
- Variables are the quantities measured in a sample.
  - Quantitative i.e. numerical
    - Continuous (e.g. pH of a sample, patient cholesterol levels)
    - Discrete (e.g. number of bacteria colonies in a culture)
  - Categorical
    - Nominal (e.g. gender, blood group)
    - Ordinal (ranked e.g. mild, moderate or severe illness). Often ordinal variables are re-coded to be quantitative.

Central Tendencies

- Center measurement is a summary measure of the overall level of a dataset
  - Commonly used methods are mean, median, mode, quantiles.
- **Mean**: Summing up all the observation and dividing by number of observations.
  - Example: Mean of 20, 30, 40 is (20+30+40)/3 = 30.

Population

$$\mu = \frac{1}{N}\sum_{j=1}^{N} x_j$$

Sample

$$\bar{x} = \frac{1}{n}\sum_{j=1}^{n} x_j$$

Central Tendencies

- **Median**: The middle value in an ordered sequence of observations. That is, to find the median we need to order the data set and then find the middle value.
  - In case of an even number of observations the average of the two middle most values is the median.
  - For example, to find the median of {9, 3, 6, 7, 5},
    - we first sort the data giving {3, 5, 6, 7, 9},
    - then choose the middle value 6.
  - If the number of observations is even, e.g., {2, 3, 5, 6, 7, 9}, then the median is the average of the two middle values from the sorted sequence, in this case, (5 + 6) / 2 = 5.5.

Central Tendencies

- **Mode:** The value that is observed most frequently. The mode is undefined for sequences in which no observation is repeated.
- **Quartiles**: Quartiles are divided into four regions that cover the total range of observed values. Cut points for these regions are known as quartiles.
  - The first quartile (Q1) is the first 25% of the data. The second quartile (Q2) is between the 25th and 50th percentage points in the data. The upper bound of Q2 is the median. The third quartile (Q3) is the 25% of the data lying between the median and the 75% cut point in the data.

Central Tendencies

- **Quartiles (cont.):**
  - Q1 is the median of the first half of the ordered observations and Q3 is the median of the second half of the ordered observations.
  - An example with 15 numbers

3 6 7 11 13 22 30 40 44 50 52 61 68 80 94

Q1 Q2 Q3

  - The first quartile is Q1=11. The second quartile is Q2=40 (This is also the Median.) The third quartile is Q3=61.

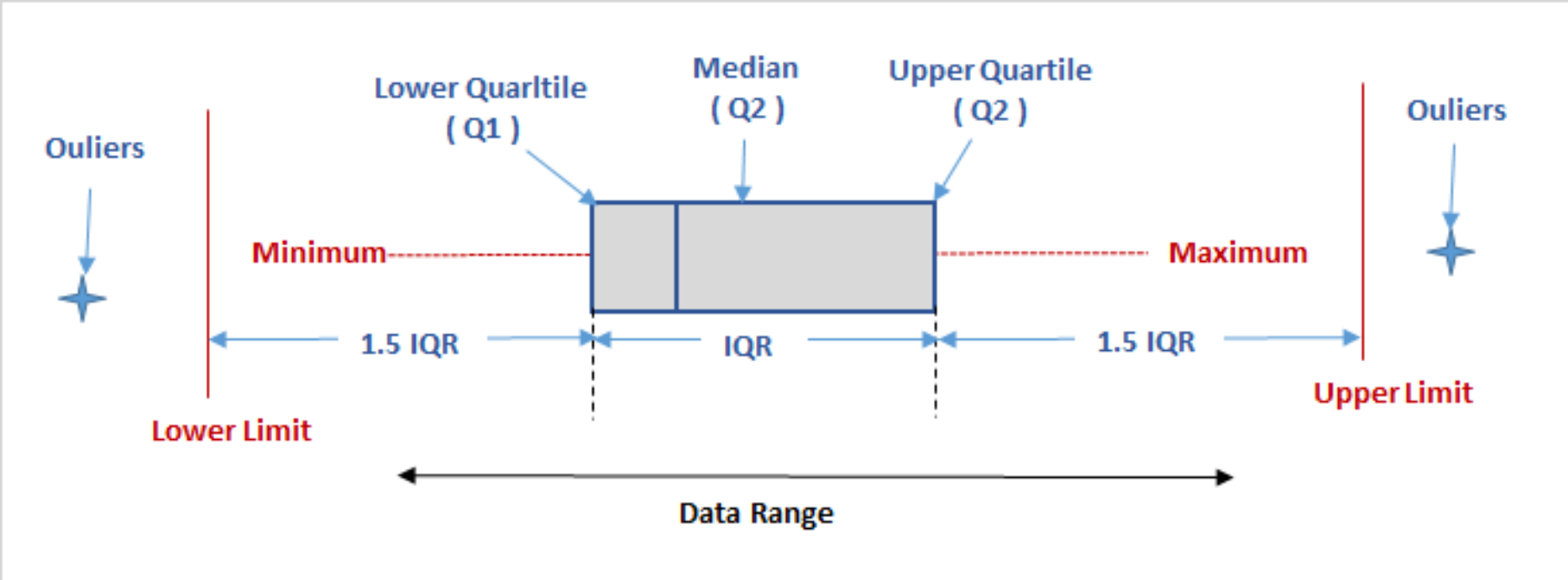
Central Tendencies

- **Deciles**: If data is ordered and divided into 10 parts, then cut points are called Deciles
- **Percentiles**: If data is ordered and divided into 100 parts, then cut points are called Percentiles. 25th percentile is the Q1, 50th percentile is the Median (Q2) and the 75th percentile of the data is Q3.
- Quantiles are cut points dividing the range of a probability distribution into continuous intervals.
- Commonly,
  - Min=quantile(0%); Q1=quantile(25%); Q2=median; Q3=quantile(75%); Max =quantile(100%)

Central Tendencies

- **Box Plot:** A box plot is a graph of the five number summary.
- The interquartile range (IQR) formula is the first quartile quartile subtracted from the third quartile: IQR = Q3 – Q1.
- **Outliers:**
  - Outliers are value that are far from the central tendency
  - Outliers might be caused by errors in collecting or processing the data, or they might be correct but unusual measurements.
  - It is always a good ideas to check for outliers, and sometimes it is useful and appropriate to discard them
  - Variance, Standard deviation, covariance, correlation can be very sensitive to outliers.

Central Tendencies



Dispersion

- Variability (or dispersion) measures of how spread out data
  - Commonly used methods: range, variance, standard deviation, coefficient of variation etc.
  - Range=max-min: The difference between the largest and the smallest observations. The range of 10, 5, 2, 100 is (100-2)=98. It's a crude measure of variability.
  - Variance: The variance of a set of observations is the average of the squares of the deviations of the observations from their mean.

Dispersion

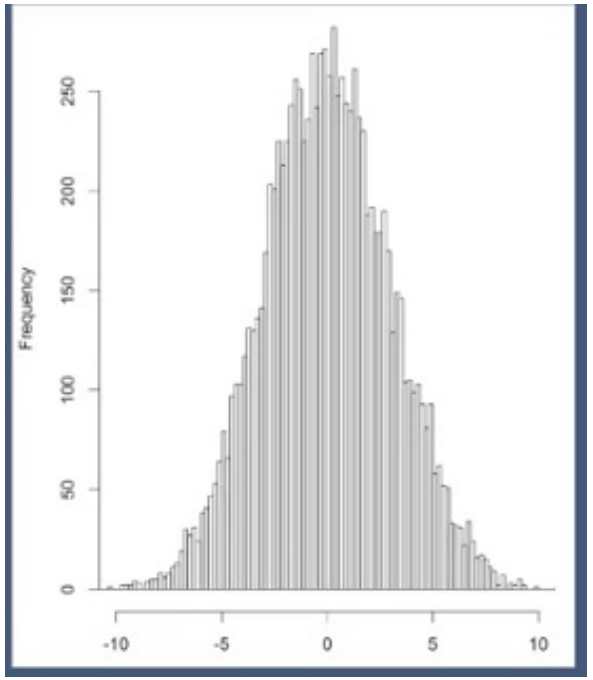
- Standard deviation (std)=square root of variance. Std measures how wide the data is spread around the mean

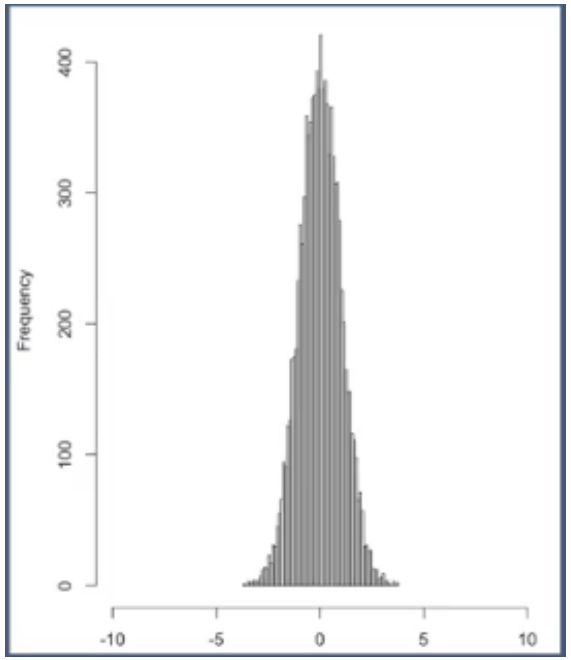
Population

$$\sigma^2 = \frac{1}{N}\sum_{j=1}^{N}(x_j - \mu)^2$$

Sample:

$$s^2 = \frac{1}{n-1}\sum_{j=1}^{n}(x_j - \bar{x})^2$$





Dispersion

- **Coefficient of Variation:** The standard deviation of data divided by it's mean. It is usually expressed in percent.

CV for a population:

$$CV = \frac{\sigma}{\mu} * 100\%$$

CV for a sample:

$$CV = \frac{s}{\bar{x}} * 100\%$$

Shape of Data

- Shape of data is measured by Skewness and Kurtosis
  - **Skewness**: How asymmetric data is spread around the mean

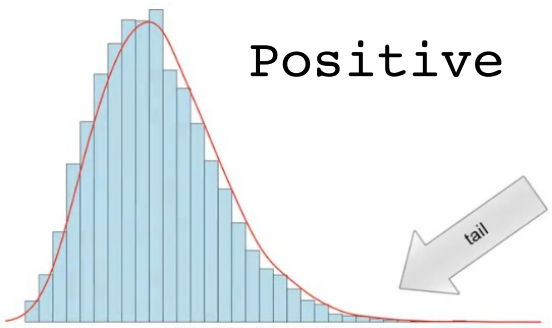
Population

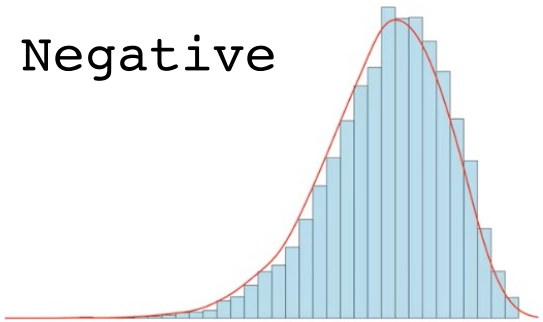
$$skewness = \frac{1}{N}\frac{\sum_{j=1}^{N}(x_j-\mu)^3}{\sigma^3}$$

Sample

$$skewness = \frac{n}{(n-1)(n-2)}\frac{\sum_{j=1}^{n}(x_j-\bar{x})^3}{s^3}$$

- Positive or right skewed: Longer right tail
- Negative or left skewed: Longer left tail
- A normal distribution has a skew of 0.





Shape of Data

- **Kurtosis** is a measure of peakedness. The higher of the kurtosis means the more outliers are present and the longer the tails of the distribution in the histogram are.
  - The kurtosis of normal distribution is 0.

Population

$$kurtosis = \frac{1}{N}\frac{\sum_{j=1}^{N}(x_j - \mu)^4}{\sigma^4}$$

Sample

$$kurtosis = \frac{n(n+1)}{(n-1)(n-2)(n-3)}\frac{\sum_{j=1}^{n}(x_j - \bar{x})^4}{s^4}$$

Covariance

- **Covariance** is a measure of the tendency of two variables to vary together

Population Covarian Formula:

$$covaviance(X, Y) = \frac{1}{N} \sum_{j=1}^{N} (x_j - \mu(X))(y_j - \mu(Y))$$

Sample Covarian Formula:

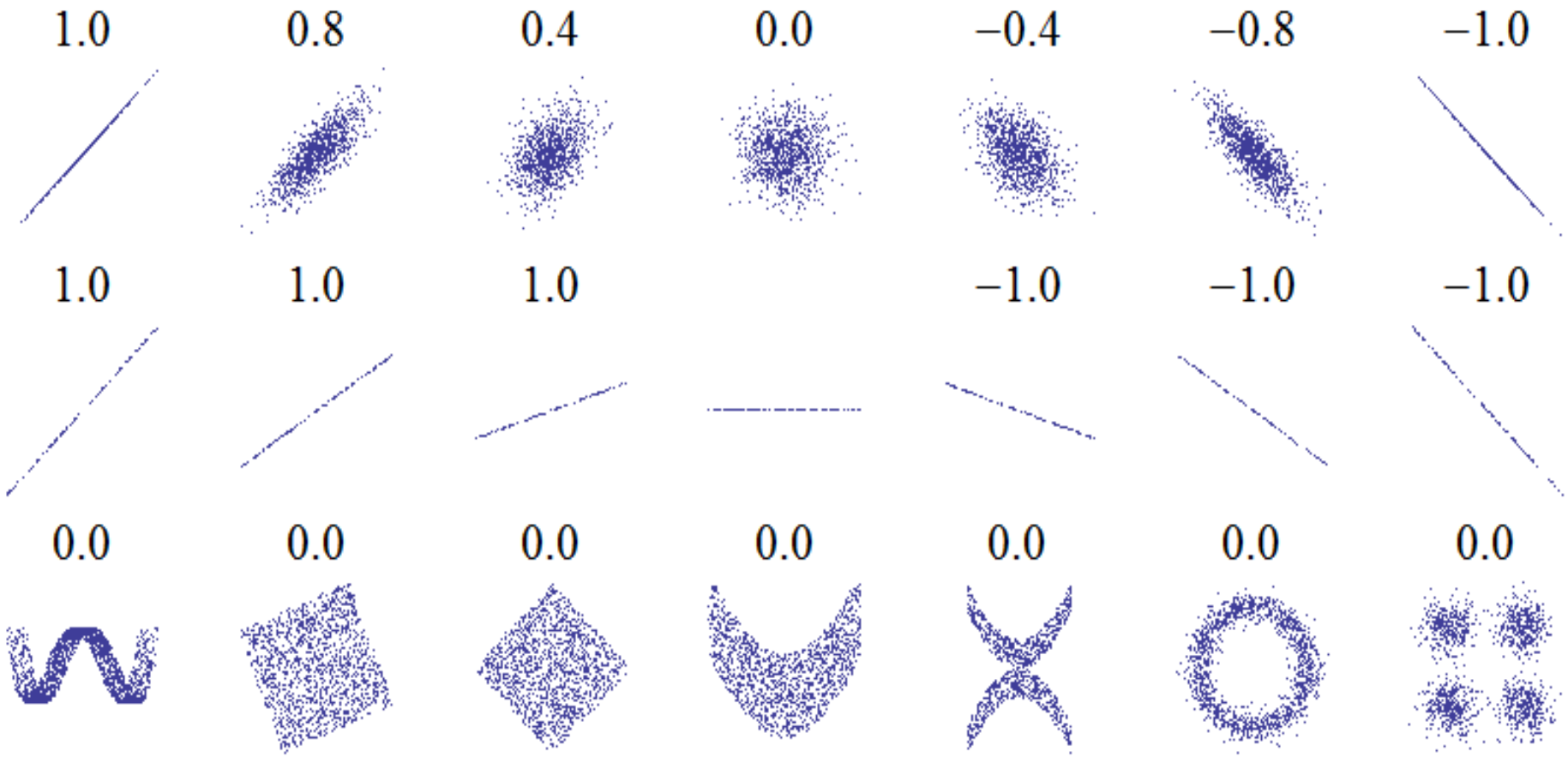
$$covariance(X, Y) = \frac{1}{n-1} \sum_{j=1}^{n} (x_j - \bar{x})(y_j - \bar{y})$$

Correlation

- **The correlation** is unitless and always lies between -1 (perfect anti-correlation) and 1 (perfect correlation). A number like 0.25 represents a relatively weak positive correlation.
- It is defined by

$$\rho = \frac{\text{covariance}(X, Y)}{\text{standard_deviation}(X) * \text{standard_deviation}(Y)}$$

Correlation example



Distribution

- The *distribution* of a statistical data set (or a population) is a listing or function showing all the possible values (or intervals) of the data and how often they occur.
  - Discrete (Frequency) distribution
  - Continuous distribution
- Example of Frequency distribution: Consider a data set of 26 children of ages 1-6 years. Then the frequency distribution of variable 'age' can be tabulated as follows:

Age	1	2	3	4	5	6
Frequency	5	3	7	5	4	2

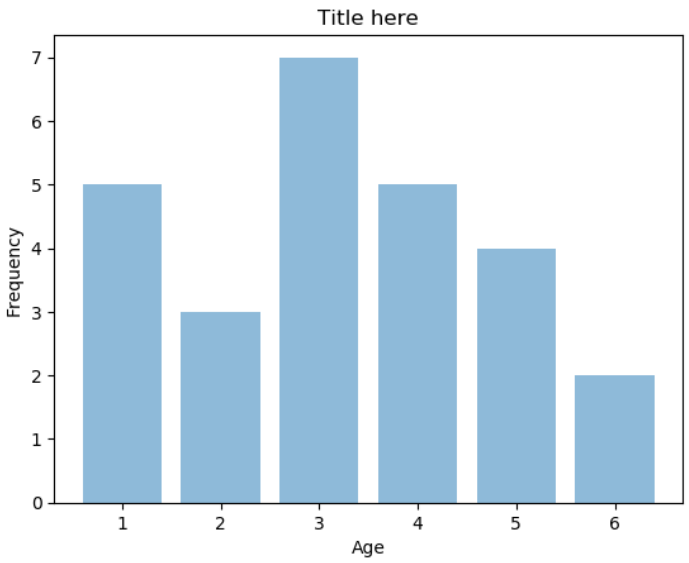
Histogram

- An accurate representation of the distribution of numerical data.
- In matplotlib (pandas also supported), seaborn is better ☺

Code

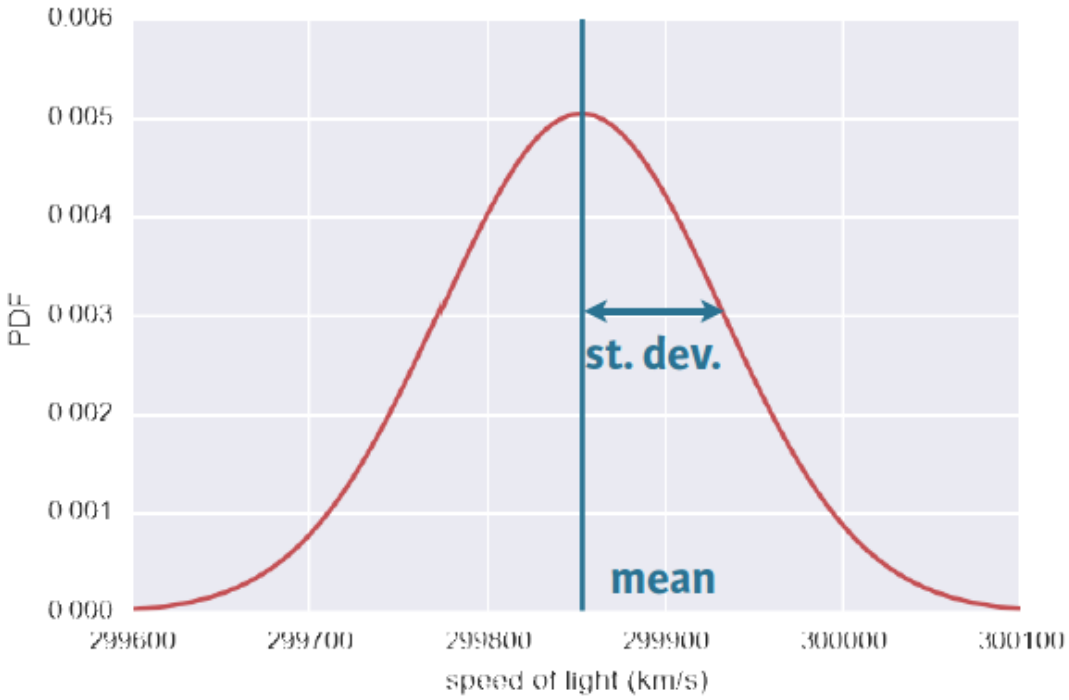
```
x = ('1', '2', '3', '4', '5', '6')
y_pos = np.arange(len(x))
x = [5,3,7,5,4,2]
plt.bar(y_pos, x, align='center', alpha=0.5)

plt.xticks(y_pos, x)
plt.ylabel('Frequency')
plt.xlabel('Age')
plt.title('Title here')

plt.show()
```

Output



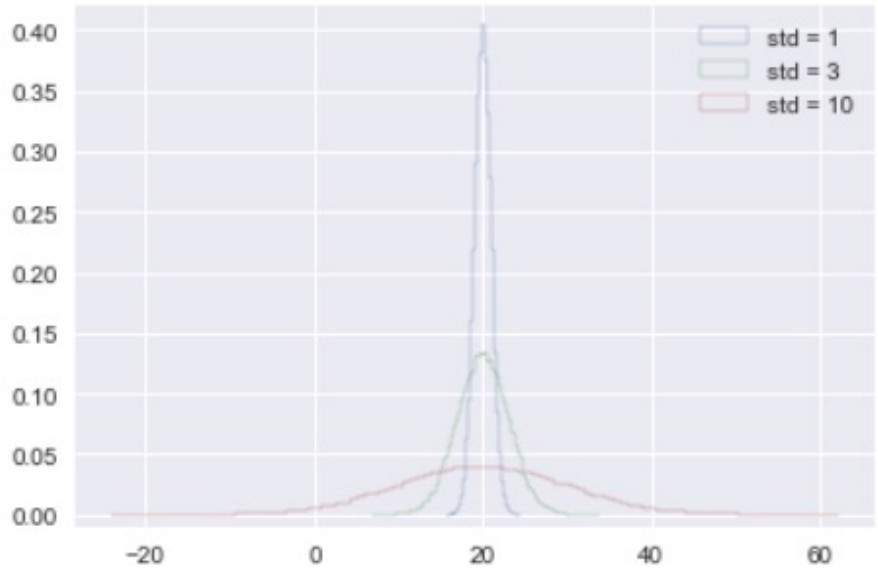
Distribution

- Normal distribution: Describes a continuous variable whose PDF has a single symmetric peak.



Distribution

```
samples_std1 = np.random.normal(20, 1, size=100000)
samples_std3 = np.random.normal(20, 3, size=100000)
samples_std10 = np.random.normal(20, 10, size=100000)

# Make histograms
_ = plt.hist(samples_std1, bins=100, normed=True, histtype='step')
_ = plt.hist(samples_std3, bins=100, normed=True, histtype='step')
_ = plt.hist(samples_std10, bins=100, normed=True, histtype='step')

# Make a legend, set limits and show plot
_ = plt.legend(('std = 1', 'std = 3', 'std = 10'))
plt.ylim(-0.01, 0.42)
plt.show()
```

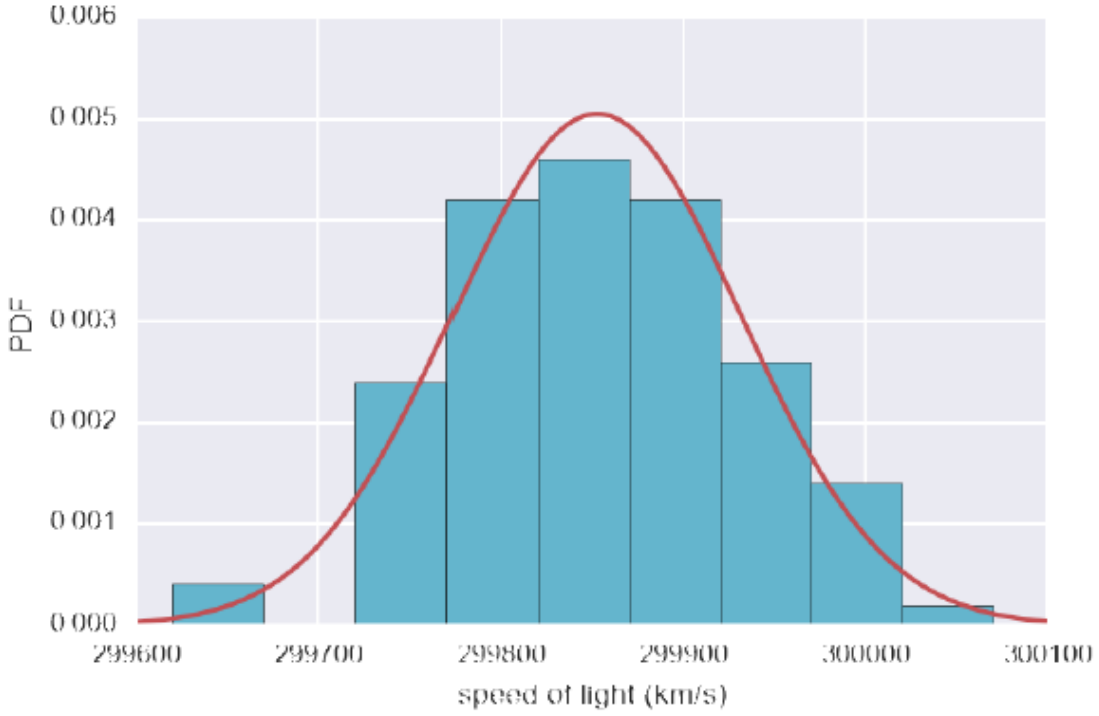


Distribution

Parameter

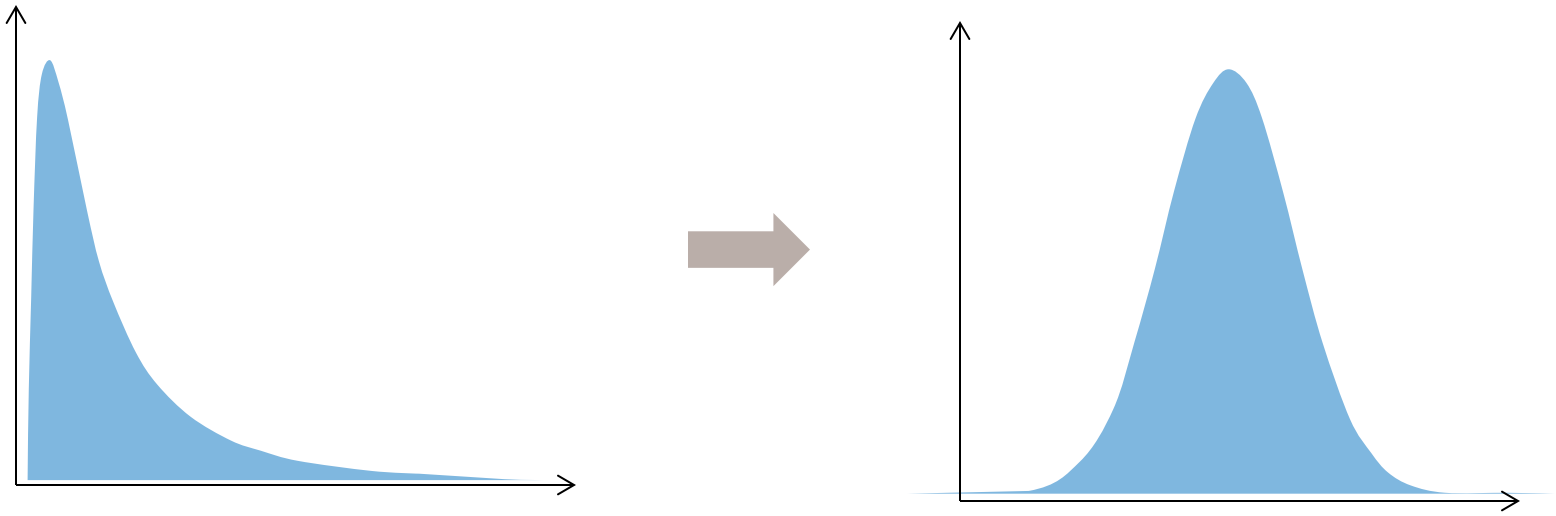
mean of a
Normal distribution

st. dev. of a
Normal distribution



Distribution

- Transformation of Data Distributions



```
from numpy import log

log_data = np.log(data)
```

Entropy

- Entropy measures the uncertainty/impurity present in the data

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

- where
  - X – set of all instances in the dataset
  - n – number of distinct class values
  - p_i – event probability

Statistical Inference

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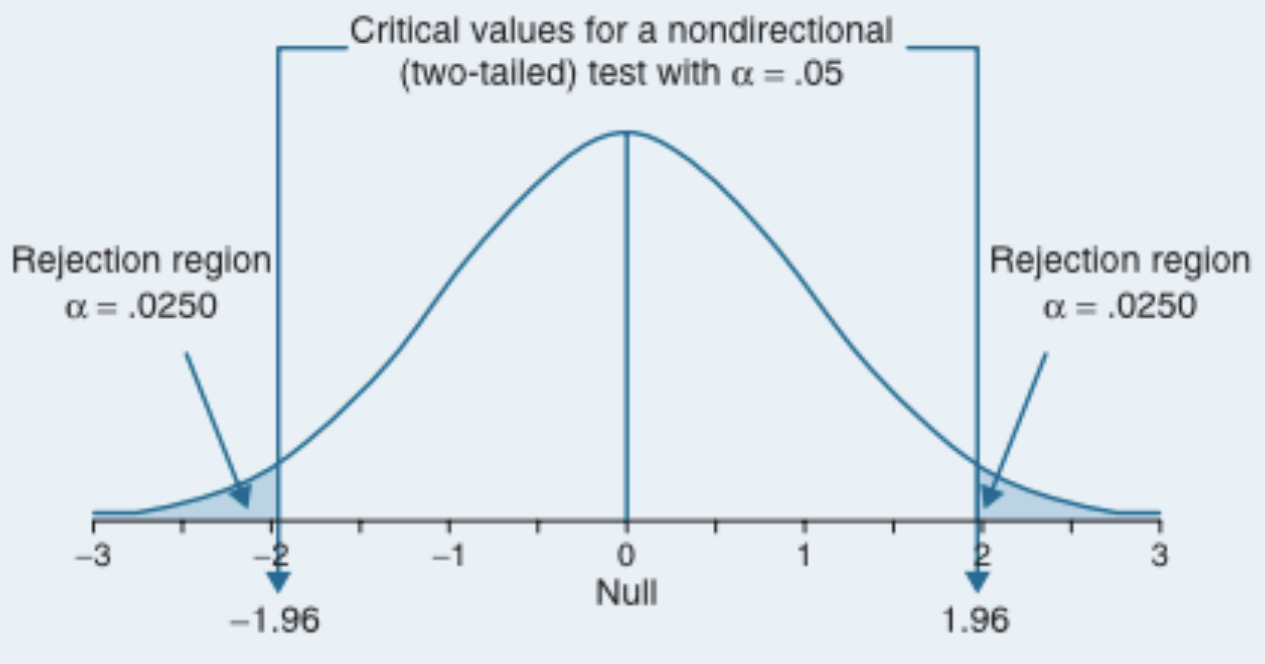
Hypothesis testing

- Hypothesis: A statement we can test, e.g., a coin is biased, groups are different,
- We want to prove a hypothesis H_1, but its hard so we try to disprove a null hypothesis H_0.
  - Alternative hypothesis H_1: Our idea, e.g., a coin is biased, groups are different,
  - Null hypothesis H_0 : The alternative of our idea, e.g., a coin is unbiased, groups are the identical.
- The threshold probability is called a *p-value*. This directly controls the false positive rate (rate at which we expect to observe large s even if is H_0 true).
- A significance level is called p_given. The common values of p-given is 0.05, 0.02, 0.01, 0.005, 0.001
- If p_value<p_given we accept H_1 (or reject H_0)

Hypothesis testing

- **Two-tailed Significance**



When the p_value is less than 5% ($p < .05$), we reject the null hypothesis

Hypothesis testing

- Pearson correlation test
- (ANOVA) F-Test
- T-Test
  - One sample
  - Two sample

Pearson correlation test

- Test association between two quantity variables. The test calculates a Pearson correlation coefficient and the p-value for testing non-correlation
- The p-value is the probability of seeing a t-statistic at least that far from 0 if the null hypothesis =“X and Y are dependent ” were true.
- Example: If X and Y are perfect correlation then p-value=1

```
import scipy.stats as stats
correlation, pvalue = stats.pearsonr(x,y)
```

F-Test

- Test 2 (or more) variances of 2 (or more) data sets are identical.
  - *p_value* is used to accept/reject the hypothesis

```
import scipy.stats as stats
f_val, p_val = stats.f_oneway(x,y,z)
```

T-Test

- One sample: test the mean of data set is equal (H_0 – null hypothesis) to a given mean.

```
import scipy.stats as stats
t_stats, p_val = stats.ttest_1samp(X, given_mean)
```

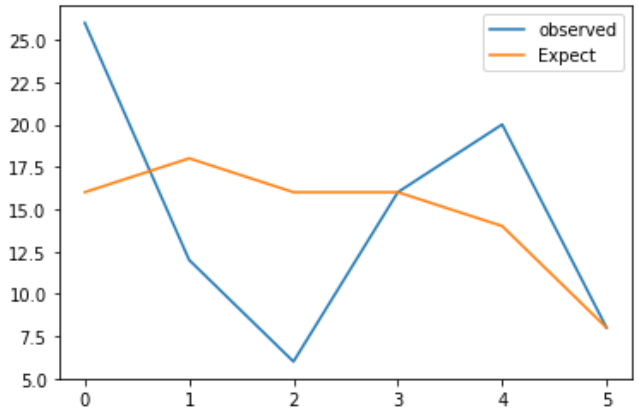
- Two sample: test two population means are equal/identical.

```
import scipy.stats as stats
T_stats, p_val = stats.ttest_ind(x,y)
```

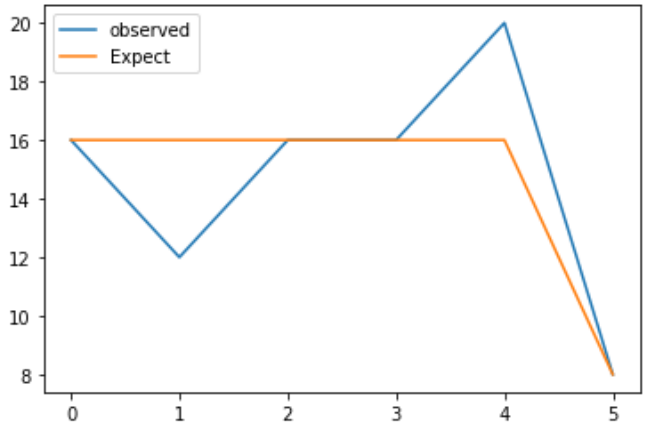
chi-square

- The chi-square test tests the null hypothesis that the categorical data has the given frequencies.
  - Chi2 small ➔ dependent
  - Chi2 large ➔ independent

```
from scipy.stats import chisquare
stats, p_value=chisquare([26, 12, 6, 16, 20, 8],
f_exp=[16, 18, 16, 16, 14, 8])
print(stats, ',', p_value) ➔ 17.071428571428573
, 0.0043663040887919075
```




```
stats, p_value=chisquare([16, 12, 16, 16, 20, 8],
f_exp=[16, 16, 16, 16, 16, 8])
print(stats, ',', p_value) ➔ 2.0 ,
0.8491450360846096
```



chi2_contingency

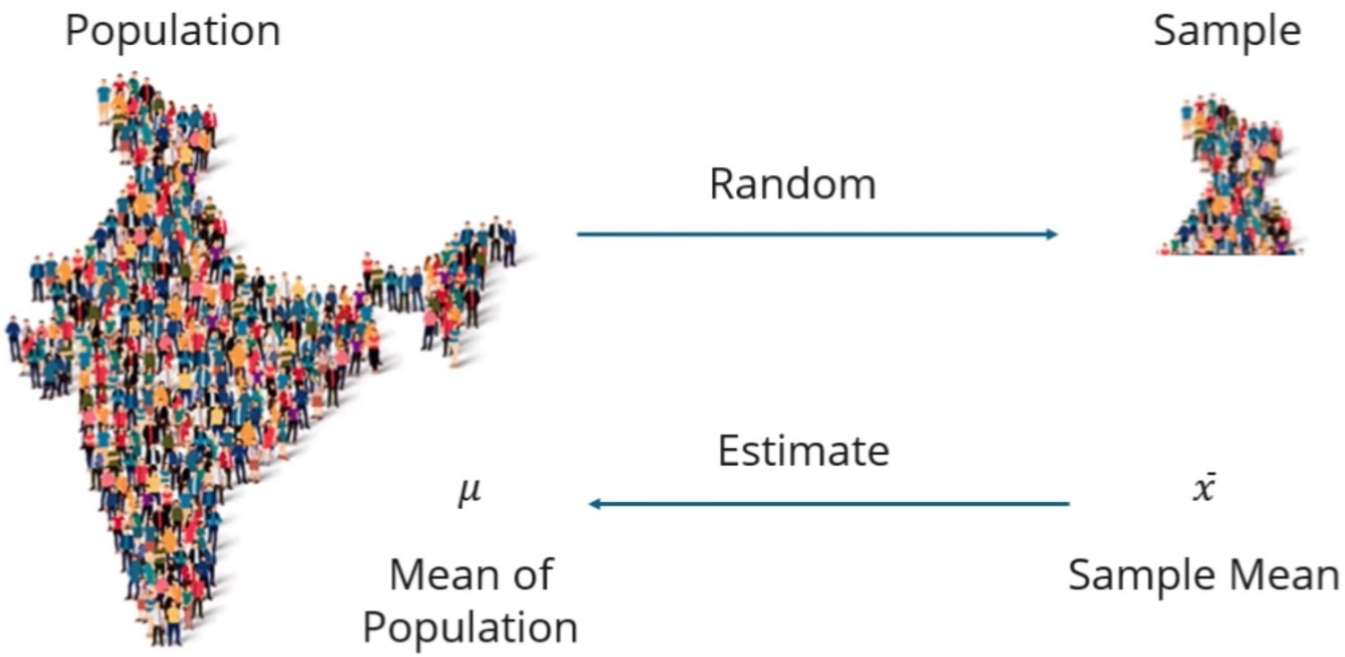
- Chi-square test of independence of variables in a contingency table.
  - In this test you are testing if there is there is relationship between two or more variable. This is called ***chi-square test for independence***, also called ***Pearson's chi-square test*** or the ***chi-square test of association***.
- Since, p-value > alpha (0.05)➔ therefore, we accept H0, that is, the variables do not have a significant relation (i.e., Independent).

```
from scipy.stats import chi2_contingency

# defining the table
data = [[207, 282, 241], [234, 242, 232]]
stat, p, dof, expected = chi2_contingency(data)
```

Inferential Statistics

- **Parameter Estimation** – concerned with the use of the sample data to measure a single value which serves as an approximate value of the best estimate of unknown population parameter.



Parameter Estimation

- **Method of Moments** – estimates are found out by equating the first k sample moments to the corresponding k population moments.
- **Maximum Likelihood Estimator** (MLE) – use a model and the value in the model to maximize a likelihood function.
- **Bayes Estimator – Maximum a Posterior** (MAP) – maximize the expectation of random variables.
- **Best unbiased estimator ???**

Maximum Likelihood Estimator

- The likelihood function is the density function regarded as a function of θ.

$$L(\theta|\boldsymbol{X}) \equiv f(\boldsymbol{X}|\theta) = \prod_{j=1}^{n} f(X_j|\,\theta)$$

- Maximum Likelihood Estimator (MLE), find θ to maximize $f(X|\theta)$
- Typically, maximizing the score function **ln L(θ|**X**)** will be easier.

Linear Regression with MLE

- Model

$$y_i = \alpha + \beta x_i + \epsilon_i$$

where the ε_i (residual) are independent mean 0 normal random variable. The unknown variance is σ^2.

The likelihood function

$$\mathbf{L}(\alpha, \beta, \sigma^2 | \mathbf{y}, \mathbf{x}) = \frac{1}{\sqrt{(2\pi\sigma^2)^n}} \exp -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

$$\ln \mathbf{L}(\alpha, \beta, \sigma^2 | \mathbf{y}, \mathbf{x}) = -\frac{n}{2} \ln 2\pi\sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

Linear Regression with MLE

- The maximum likelihood estimator for σ^2

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{k=1}^{n}(y_i - \hat{y}_i)^2.$$

- The maximum likelihood estimators $\hat{\alpha}$ and $\hat{\beta}$ also the least square estimator.

$$\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i.$$

Bayes Estimator – Maximum a Posterior

- Look back Bayes Theorem with θ be parameter model and D be data.

"Posterior" "Likelihood" "Prior"

$$P(\theta \mid D) = \frac{P(D \mid \theta)\, P(\theta)}{P(D)}$$

- We have prior $P(\theta)$ and can compute $P(D \mid \theta)$
- But how do we calculate $P(D)$?
  - It complicated, but not depend on θ, ➔ ignored it (just constant).

Maximum a Posterior

- Maximum A Posteriori (MAP) estimator of θ:

$$\theta_{MAP} = \arg\max_{\theta} f(\theta \mid X_1, X_2, \ldots, X_n) = \arg\max_{\theta} \frac{f(X_1, X_2, \ldots, X_n \mid \theta)\, g(\theta)}{h(X_1, X_2, \ldots, X_n)}$$

$$= \arg\max_{\theta} \frac{\left(\prod_{i=1}^{n} f(X_i \mid \theta)\right) g(\theta)}{h(X_1, X_2, \ldots, X_n)} = \arg\max_{\theta} g(\theta) \prod_{i=1}^{n} f(X_i \mid \theta)$$

  - $g(\theta)$ is prior distribution of θ.
  - $f(X_1, X_2, \ldots, X_n \mid \theta)$ is likelihood function.

Maximum a Posterior

- As before, can often be more convenient to use log:

Estimated parameter

Log prior

$$\theta_{\text{MAP}} = \underset{\theta}{\operatorname{argmax}} \left(\log(g(\theta)) + \sum_{i=1}^{n} \log(f(X_i|\theta)) \right)$$

Chose the value of theta that maximizes:

Sum of log likelihood